AN OBLIVIOUS SUBSPACE INJECTION NEED NOT GIVE RELATIVE-ERROR SKETCH-AND-SOLVE

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ABSTRACT. This note records a statement refuted in the *Count ex Machina* archive: each is quoted as posed and answered by an explicit witness, and the archive carries a certificate that recomputes the refutation by exact arithmetic.

Summary: A one-dimensional oblivious subspace injection with injectivity 1 can make sketch-and-solve incur the fixed factor $\sqrt{2}$ with probability 1/50. *Certificate:* Three exact rationally weighted 2×2 sketch atoms, with exact isotropy, an exact 99/100 injection guarantee, and exact squared residual ratios.

Posed by Amsel et al. (2026); counterexample found by OpenAI Codex (2026-08); formalized by OpenAI Codex; audited by OpenAI Codex; contributed by Suvrit Sra.

For a random matrix $\Omega \in \mathbb{R}^{n\times k}$ and a fixed r-dimensional subspace $\mathcal{V} \subseteq \mathbb{R}^n$, the source calls Ω a subspace injection with injectivity α on $\mathcal{V}$ when $\mathbb{E}\|\Omega^T x\|_2^2 = \|x\|_2^2$ for every $x \in \mathbb{R}^n$ and $\|\Omega^T x\|_2^2 \geq \alpha\|x\|_2^2$ for every $x \in \mathcal{V}$. It is oblivious with failure probability δ if this lower inequality holds on every fixed r-dimensional subspace with probability at least $1 - \delta$. When no failure probability is displayed, the source fixes a small constant, specifically $\delta = 0.01$ as its example. The discussion immediately following Problems 5.1–5.2 asks whether constant OSI failure probability suffices and then separately asks whether reducing it to poly(ε) rescues the result. Thus the probabilistic claim tested here is the natural one in that context: the relative-error event should hold with at least the OSI success probability. Townsend and Wang subsequently gave related counterexamples and resolved the question negatively (Townsend and Wang, 2026).

Problem 1 (As posed: Problem 5.1 of the Simons workshop report of Amsel et al. (Amsel et al., 2026), scribed for Section 5 by Raphael A. Meyer. Townsend and Wang (Townsend and Wang, 2026) had already resolved the general question; this witness is independent and claims no priority.)**.** Let $A \in \mathbb{R}^{n\times d}$ be a full-rank matrix, $B \in \mathbb{R}^{n\times p}$ be a matrix, and let $\Omega \in \mathbb{R}^{n\times k}$ be an oblivious subspace injection with dimension d and injectivity $1 - \varepsilon$. Let $\widetilde{X} = (\Omega^\top A)^+(\Omega^\top B) \in \mathbb{R}^{d\times p}$ be the sketch-and-solve approximation to the ordinary least-squares solution $X = A^+B$. (Here, $^+$ denotes the Moore–Penrose pseudoinverse.) Prove or disprove: Does it necessarily hold that

$$\|A\widetilde{X} - B\|_{\mathrm{F}} \leq (1 + \mathcal{O}(\varepsilon)) \min_{X\in\mathbb{R}^{d\times p}} \|AX - B\|_{\mathrm{F}}?$$

Problem 1 asks whether one-sided injectivity near 1 yields the usual relative-error guarantee for sketch-and-solve (Amsel et al., 2026). The following three-atom distribution gives a negative answer under the failure probability convention fixed in the source.

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From the *Count ex Machina* archive of counterexamples, https://github.com/suvrit/count-ex-machina, whose companion paper is arXiv:2608.29595.

Theorem 2 ([**Disproof**]: OSI relative error for sketch-and-solve)**.** *There is a* $(1, 1, 1/100)$ *oblivious subspace injection for which the claimed relative-error inequality fails with probability* $1/50$*. Consequently no* $1 + O(\varepsilon)$ *factor can hold with the OSI success probability, even though the injectivity is exactly* 1*.*

Proof. Put $\rho = 1/100$ and define

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad B_+ = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \qquad B_- = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}.$$

Let Ω equal I, B_+, B_- with respective probabilities $1 - 2\rho, \rho, \rho$. Its isotropy is exact:

$$\mathbb{E}[\Omega\Omega^T] = (1 - 2\rho)I + \rho \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \rho \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = I.$$

For $v = (x, y)^T$, the three possible values of $\|\Omega^T v\|_2^2$ are

$$x^2 + y^2, \qquad (x + y)^2, \qquad (x - y)^2.$$

The identity $(x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$ shows that at least one of the last two values is at least $x^2 + y^2$. Hence, on every fixed line, the injectivity-1 inequality can fail on at most one of the two atoms B_+ and B_-, each of probability ρ. Thus Ω is a $(1, 1, \rho)$-OSI, and in particular a $(1, 1 - \varepsilon, \rho)$-OSI for every $\varepsilon \in (0, 1)$.

Now take the single-response problem

$$A = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad b = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The unsketched optimum is $x_\star = 0$ and its residual norm is 1. On the identity atom, $\widetilde{x} = 0$. On the other atoms,

$$B_+^T A = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad B_+^T b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad B_-^T A = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad B_-^T b = \begin{pmatrix} -1 \\ 0 \end{pmatrix}.$$

Therefore $\widetilde{x} = 1$ on B_+ and $\widetilde{x} = -1$ on B_-, and in both cases

$$\|A\widetilde{x} - b\|_2^2 = \widetilde{x}^2 + 1 = 2.$$

The fixed loss $\sqrt{2}$ occurs with total probability $2\rho = 1/50$, whereas the OSI failure probability is only $\rho = 1/100$.

Finally, this is genuinely an asymptotic refutation of $1 + O(\varepsilon)$. If a universal constant $C > 0$ supplied the hidden constant, choose $\varepsilon > 0$ so small that $C\varepsilon \le 1/4$. The same sketch still has injectivity at least $1 - \varepsilon$, while on an event of probability $1/50$ its ratio is $\sqrt{2} > 5/4 \ge 1 + C\varepsilon$. No rounding or numerical approximation is used. □

Code and certificate

The case is the directory `counterexamples/osi-sketch-and-solve/` of the archive, at commit `7240442`. Its `verify.py` recomputes every refutation in this note using exact arithmetic or rigorous interval enclosures, never floating-point evidence alone; it raises an error if any check fails and otherwise writes the certificate below. To rerun it:

```
git clone https://github.com/suvrit/count-ex-machina
cd count-ex-machina/counterexamples/osi-sketch-and-solve
python verify.py
```

- Theorem 2 (result `osi-sketch-and-solve`): exact arithmetic.
- `artifacts/certificate.json`: Exact sketch atoms, probabilities, isotropy matrix, least-squares solutions, and residual ratios

AI USAGE

The credit line under the title records, for each statement, who posed it, who found its counterexample and when, and who formalized and audited it; the counterexample here was found by OpenAI Codex (2026-08). This note was assembled mechanically from the case's source in the archive by `tools/notes.py`, written with Claude Code; its mathematics is exactly that of the archive at the commit cited above.

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